Lesson Plan: $+C$

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0.1 Introduction

In an introductory course on single-variable calculus, students learn to evaluate integrals by using a handful of techniques to compute the anti-derivative and then adding an arbitrary constant of integration: $+C$. It is common for a student to not understand why or when to add $+C$, and for this issue to escalate in later courses on calculus and differential equations.

Herein, we offer an explanation to a student of integral calculus who is willing to learn (or believe) a few "advanced" tricks.

0.2 Exploration

In this section, we will present some high-level definitions and technical lemmas from linear algebra, prove a theorem from ODEs, and present the calculus result as a corollary.

0.2.1 Differential Equations

Let $n \in \mathbb{N}$, and let $C^k = C^k_M(\Omega)$ be the vector space over a manifold M of continuous real-valued functions on a connected submanifold $\Omega \subset M$ which possess $k \in \mathbb{N}$ continuous derivatives. Let $\Omega = (t_0, t_1)$ be an open interval of real numbers $t_0, t_1 \in M = \mathbb{R}$.

Definition 0.2.1 (n^{th}-order linear ordinary differential operators)**.** Consider the vector subspace of $\text{Hom}_{\text{Vect}_{C^0}}\left(C^{n+1} \to C^0\right)$ which consists of the following linear transformations:

$$\varphi = \varphi_q : y \mapsto \sum_{k=0}^{n} q_k(t) y^{(k)}(t), \qquad \forall\ q = (q_0, q_1, \ldots, q_{n-1}, q_n) \in \left(C^0\right)^n \times C^0_\star. \tag{1}$$

We say φ is monic if $q_n(t) = 1$ everywhere.

Definition 0.2.2 (non-homogeneous ODE)**.** An nth-order, linear nonhomogenous ordinary differential equation of an unknown function $y \in C^{n+1}$ is an equation of the following form:

$$\varphi(y) = g, \tag{2}$$

where we seek the space $\varphi^{-1}(g)$ of non-homogeneous solutions for a given function $g \in C^0$. Such equations have monic representatives, obtained by dividing $g(t)$ and $p(t)$ (component-wise) on domains where they are well-defined.

Definition 0.2.3 (homogeneity)**.** An ODE is said to be homogeneous if $g(t) = 0$ everywhere, i.e. it is of the following form:

$$\varphi(y_H) = 0, \qquad y_H \in \varphi^{-1}(0). \tag{3}$$

Lemma 1. *The homogeneous solutions form an n-dimensional vector subspace:*

$$\varphi^{-1}(0) = \ker(\varphi) \subset C^{n+1}. \tag{4}$$

Thus, it is spanned by some fundamental set $\{y_k\}_{k=1}^n$ of solutions, i.e.:

$$\forall y_H \in \ker(\varphi), \qquad y_H(t) = \sum_{k=1}^{n} c_k y_k(t), \qquad \textit{for some } c_k \in \mathbb{R}. \tag{5}$$

Proof (Hernandez). This follows directly from the first isomorphism theorem (of vector spaces). □

Suppose $Y_1(t)$ and $Y_2(t)$ satisfy the non-homogeneous Equation 2, and consider $Y(t) = Y_1(t) - Y_2(t)$.

Lemma 2. *There exist constants $c_k \in \mathbb{R}$ such that $Y(t) = \sum_{k=1}^{n} c_k y_k(t)$.*

Proof (Hernandez). For all $t \in \Omega$, we obtain the following equation:

$$\begin{aligned}
\varphi(Y) = \varphi(Y_1 - Y_2) &= \sum_{k=0}^{n} q_k(t)(Y_1 - Y_2)^{(k)}(t) && \text{by substitution} \\
&= \sum_{k=0}^{n} q_k(t)(Y_1^{(k)}(t) - Y_2^{(k)}(t)) && \text{by linearity} \\
&= \varphi(Y_1) - \varphi(Y_2) = g(t) - g(t) = 0 && \text{by associativity and hypothesis.}
\end{aligned}$$

Thus, $Y(t)$ is a homogeneous solution. Our result now follows from Lemma 1. □

Alternatively, we could have proven this theorem by citing the fact that any vector can be constructed as the sum of a particular vector in the row space and a vector in the null-space.

Theorem 3. *Any non-homogeneous solution can be expressed as the sum of a homogeneous solution and a known particular non-homogeneous solution as shown below:*

$$y(t) = Y_P(t) + y_H(t). \tag{6}$$

Proof (Hernandez). Let $y(t)$ be the general solution to Equation 2, and suppose $Y_P(t)$ and y_H are known. Then, our result follows by applying Theorem 2 to $Y = y(t) - Y_P(t)$ and adding $Y_P(t)$ to both sides of the equation. □

In this way, solving the non-homogeneous equation $y(t)$ amounts to the much easier tasks of finding the general homogeneous solution $y_H(t)$ and finding some particular solution $Y_P(t)$.

0.2.2 Integral Calculus

Consider a first-order, linear nonhomogenous differential equation and its associated homogenous differential equation:

$$y'(t) = g(t), \quad y'_H(t) = 0, \qquad \forall t \in \Omega, \tag{7}$$

as in Equation 2 with $n = 1$, $q_0(t) = 0$, and $q_1(t) = 1$. Given $g(t)$, we seek a function $y(t)$ which satisfies the following equivalent equation:

$$y(t) = \int \frac{dy}{dt}\, dt = \int g(t)\, dt, . \tag{8}$$

which is a simplification obtained by anti-differentiating Equation 7.

Corollary 4. *The general solution to an integral equation can be expressed as follows:*

$$y(t) = \int_0^t g(s)\, ds + c. \tag{9}$$

Proof (Hernandez). As in the previous section, we obtain the following equation:

$$y(t) = \int g(t)\ dt = Y_P(t) + y_H(t) = \int_0^t g(s)\ ds + c,$$

where $Y_P(t) = \int_0^t g(s)\ ds$ can be computed symbolically through various integration techniques and where $y_H(t) = c_1 y_1(t)$ is a scalar multiple of a homogeneous solution $y_1(t)$. Notice $y_H'(t) = 0$ can be satisfied by $y_1(t) = 1$ or any other constant function $y_1 \in \mathbb{R}$.

We check this result by applying the first fundamental theorem of calculus as follows:

$$\frac{d}{dt}(y) = \frac{d}{dt}\left(\int_a^t g(s)\ ds + c\right) = \frac{d}{dt}\left(\int_a^t g(s)\ ds\right) + \frac{d}{dt}(c) = g(t) + 0 = g(t). \tag{10}$$

□

This is precisely where the constant $+c$ of integration comes from.

0.3 Summary

We leveraged linear algebra to prove a famous theorem from ODEs and thus derive the constant $+c$ of integration from first principles, and then checked the result as it is frequently taught.

Although the approach may seem quite foreign or elaborate in this context, it serves to motivate similar techniques which appear naturally in further courses on linear algebra, partial differential equations, and differential geometry.